Creating Depth

This draft is mainly to start adding depth to sections

Introduction

AI has become one of the hottest topics in our modern society. The capabilities of such a complex system that is able to do tasks that would typically require a human being to do are far beyond what we have ever thought possible, and are still growing at this very moment. And one may ask, what is AI doing right now? A major focus at the moment is AI and its use in workflows.

AI is something that most people are now becoming familiar with as it has begun to impact lives.

AI, what is it and what is it for

AI, otherwise known as artificial intelligence, is a system that takes algorithms and progressively learns from those algorithms. AI is used to further our own knowledge and understanding of the world as we know it. AI also is used for completing tasks that would take serious human input and performance in a much shorter time with only. Most AI, in this case using LLMs (Large Language Models) as an example, use parameters for their main function. Parameters are the AI equivalent of the synapses in the brains of living beings. The more parameters that the AI has, the more it can do. These

parameters help it to do, they allow memorization and can even understand nuances in speech and at he highest degree, it is capable of abstract thinking that was originally thought to be uniquely human. AI has a plethora of uses, the main one in the workforces is automation. We have been able to begin using AI for its ability to automatically complete tasks that it is given at a much more rapid pace than humans can. In essence, the tasks that usually require human judgment and reasoning are now able to be completed by a machine at paces that far exceed our own capacities. The type of AI that we are presently working with is called narrow AI. Narrow AI is used for specialized and specific tasks and requires large amounts of training data. Since narrow AI can only work within certain bounds it is not capable of independent activity or “thought”. Think of any AI that we use every day like ChatGPT, it can answer questions, take large amounts of information and summarize them, and even do some straightforward processes with the right training. Currently, narrow AI is at the front when it comes to AI in business and it is being used in repetitive tasks to increase productivity and reduce costs. The next step for AI that we are aiming for is general AI, which is a massive improvement over the current narrow that we are working with. General AI in theory is a machine that is comparable to human intelligence. These machines would be fully capable of independent learning without being restricted to any specific area or function. This form of AI is the type that many would think of in science fiction, the kind that would be able to transfer information between completely unrelated tasks and solve problems without even needing to be trained in those fields to work on them. What truly would set general AI apart from narrow would be its ability to take into account nuances and fine details and be able to work them into its process, able to produce feats such as

diagnosing illnesses with incredible precision or even learn completely new skills independent from human intervention. As to why this has not been achieved yet, researchers have a few areas that they are struggling with.

History of AI

Firstly, how can one even begion to talk about AI without mentioning Alan Turing and the Turing test. Alan Turing designed what was know at the time as “the Turing machine”. The centerpiece of the machine is a strip of tape, divided into individual boxes. Each box contains a symbol, the machine using A,C,T, G based on the letters of genetic code, or a blank space. The strip of tape is the same as today’s hard drives that store data. Initially, the string of symbols on the tape corresponds to the input, containing the data for the problem to be solved. The string also serves as the memory of the computer. The Turing machine writes onto the tape the data that it needs to access later in the computation. The reason the Turing machine was so impactful was that it was the first time a true computer was created as an answer to many mathemeticians question of if there was a machine or machine that could compute all problems that were able to be computed.This machine was the very first, real computer and it was the origin point of the computers that we use today. Narrow AI is everywhere, even in places we may not even think about. It’s in our smartphones for facial recognition unlock, it’s in our online maps that figure out the fastest routes with the least traffic to our destination and we take it for granted every single day without even realizing it.

How does it work

AI takes data and processes it, before using algorithms to find patterns and make decisions based on its training. The training is how AI starts learning, and is based off of what the initial data it is given is. The creators of each AI use things called weights and biases to mimic the human brain and its ability to remove inefficient brain synapses. Every single prompt that the AI is given is sent throught these weights and biases and each of them have a pacific role. Weights are used to helop the AI determine how important the relation of one word to another is, like the word “bank” may be closely associated with the words “river” or “money”. As for biases, they are used to help the AI with its final outcome or “decision” so that the model can better match the patterns it has learned from

What are workflows

Workflows are step-by-step plans for individual tasks to be done in a certain order for optimal results. A workflow is often created by a project manager who needs to understand the project and how it needs to be done. The manager will then assign certain tasks to be done in an order that best suits the project and the people workion on it.

Why are they important

Without workflows, projects would likely be in complete dissarray. Not only do they Diffuse the workload throughout multiple people and even departments, they also give an estimated time of completion for the stakeholders and higher ups. Workflows can also find and accurately assess any sort of roadblocks within the project which may require it to be completely reset or stopped altogether. This can help save the project from wasting resources if the problems are identified early enough.

How does AI interact with workflows

Ai can be treated like an extra pair of hands in a project. This can exponentially speed up the process of project workflows by offloading some of the simpler and often more tedious work to the AI while the main workforce tackle the larger and more complex tasks. The further AI is developed, the more tasks it will be able to handle, possible even the more difficult ones that requires a human touch.

Chaining_tasks_ai_automation MIT research paper

What Is a Workflow? | IBM

AI and LLM Parameters Explained: From Millions to Trillions (2026 Guide)

Alan Turing's Everlasting Contributions to Computing, AI and Cryptography | NIST

Journal/Publishing format:

publishingportal.ieee.org/app/publication-venue?id=J-8566059